

EMILY DEEB

Spatial Data Scientist | Urban Analytics | MSc Urban Spatial Science, CASA UCL
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SUMMARY

I help cities and organisations make evidence-based spatial decisions using machine learning and GIS on urban-scale data. Over the past five years, I have led design and geo-computational workflows for municipal and private clients, developing reproducible data pipelines that increased team efficiency by 30% and translating complex spatial analyses into recommendations for planning authorities and practical design solutions. With a postgraduate certificate in Urban Spatial Science at CASA UCL, carrying out an industry dissertation with CACI Ltd in which I model national retail spending patterns using Python, spatial autocorrelation, and predictive analytics. I intend to combine behavioural analytics with tangible spatial solutions in urban, commercial, and policy settings.

EXPERIENCE

CACI Ltd – Dissertation Placement – London, UK

May 2026 – Aug 2026

- Industry-sponsored dissertation examining whether urban morphological fabric type predicts retail spending patterns across UK cities, independently of demographic composition.
- Built Python-based machine learning pipeline using GeoPandas, scikit-learn, SHAP, and spatial autocorrelation analysis to model retail spending patterns at national scale, integrating CACI consumer spending data with HiMoC morphometric classifications.
- Communicated findings through data visualisation and written reporting to industry stakeholders; presented to CACI's Consumer and location Intelligence team.

Buro Happold – Consultancy Project – London, UK

Jun 2026

- Selected through competitive UCL application to deliver a two-week industry consultancy project on smart campus strategy, working directly with Buro Happold's Digital Advisory team.
- Applied urban spatial analysis and digital strategy skills in a live multidisciplinary consultancy context; final outputs shared with Buro Happold's full UK Advisory team.

Derman Verbakel Architecture – Team Leader – Urban Design & GIS – Tel Aviv, Israel

2020 – 2025

- Led a team of 4 architects across 15 concurrent projects spanning block to city scale; strategic masterplans, TOD, urban renewal, and housing schemes. For municipal and private clients including Eilat Municipality and the Israeli Authority for Urban Renewal.
- Developed geo-computational workflows in QGIS and ArcGIS for morphological mapping, site feasibility analysis, and profitability modelling; improved office workflow efficiency by 30%.
- Collected, cleaned, and aggregated spatial datasets from scratch for historical city centre analysis, produced evidence-based spatial reports and cartographic visualisations presented to municipal planning committees and statutory authorities.
- Trained 15+ team members in QGIS and ArcGIS; developed reproducible, documented spatial workflows adopted across the office.
- Key projects: Transportation Masterplan of Eilat; Regeneration of Historical City Centre of Sakhnin (1,200 units); TOD Regeneration Plan, Rishon LeTsiyon (3,000 units); City expansion masterplan (100,000 units, confidential).

SYL Studio Yigal Levi Architecture – Architect Intern – Jerusalem, Israel

2019 – 2020

- Gained foundational experience in architectural design, documentation, and site coordination.

ACADEMIC EXPERIENCE

Bezalel Academy of Arts and Design – Professor Assistant – Jerusalem, Israel

2023 – 2025

- Co-taught second-year B.Arch design studio, mentoring twenty students through design critique and spatial analysis workshops.

SKILLS

Spatial Analysis: QGIS, ArcGIS, GeoPandas, Spatial Network Analysis, Spatial Data Science, Google Earth Engine API

Programming: Python (Pandas, GeoPandas, NumPy, Matplotlib, Scikit-learn), R (RStudio), SQL, React
Machine Learning: Classification, Clustering, Regression, SHAP, Spatial Autocorrelation, Pre-modelling Diagnostics

Data & Viz: Data Pipelines, Reproducible Workflows, Data Visualisation, Power BI, Tableau, Quarto, Jupyter

Urban & GIS: Morphometric Analysis, Network Analysis, Block/Plot Typology, Transport Modelling

Other: Git, AutoCAD, Rhinoceros, Adobe Suite, Microsoft Office, UI/UX (CalArts certified)

EDUCATION

CASA, University College London

MSc Urban Spatial Science, 2025 – 2026

- Specialising in computational urban analysis, spatial data science, geocomputation, and machine learning for cities.
- Core Modules: GIS & Science | Quantitative Methods | Urban Systems Theory | Dissertation.
- Optional modules: Data Science for Spatial Systems | Building Spatial Apps with Big Data | Urban Data Visualisation | Urban Simulation.

Bezalel Academy of Arts and Design, Jerusalem

B.Arch Architecture, 2015 – 2020

- Graduation Project: Destination: Jericho — research-led urban design study using QGIS, spatial analysis, and data visualisation.
- Exchange: Infrastructure Studio, Universität der Künste Berlin (2018). Summer Studio: XL Travelogues, Tokyo (2017).

LANGUAGES & CERTIFICATIONS

Arabic (native) | English (fluent, C1) | Hebrew (fluent)

Licensed Architect — Israeli Labour Department, 2025 | UI/UX Design — CalArts certified